

Objective

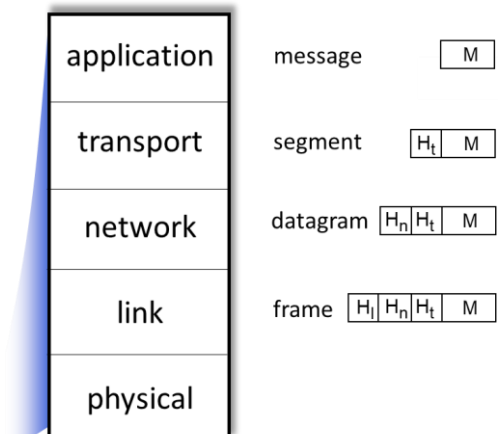
Q1. 26 marks

(a) 24 marks

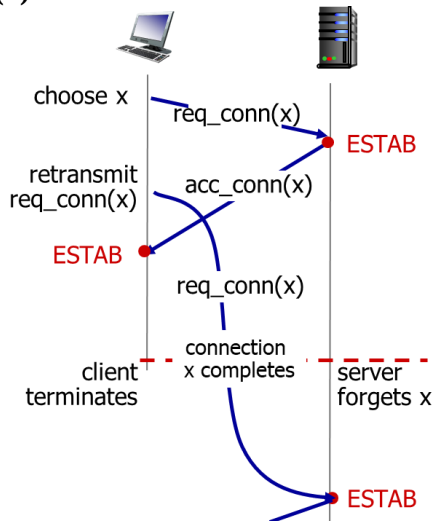
- i. Glass
- ii. Reflection, obstruction by objects, Interference/noise
- iii. Because more channels (frequency bands) are allocated for download/downlink, than for upload/uplink.
- iv. WiFi (WLAN)
- v. Fiber Optic Cable.
- vi. If the signal strength of another signal is under tolerable range, it is noise, else interference.
- vii. 16 bit, or 64K.
- viii. DNS uses UDP so the request or response can be lost.
- ix. The TCP will immediately sends cumulative ACK of the two segments.
- x. No, UDP does not perform retransmissions.
- xi. The rdt_send() function is invoked by application layer to inform transport layer, that an application wants to send some data to the specified receiver(s).
- xii. No parameter is passed in this function.
- xiii. TCP Reno have Fast Recovery Phase, while TCP Tahoe don't have
TCP Reno cut window size in half on triple duplicate ACK, while TCP Tahoe window size always start from 1 MSS.
- xiv. 16 bit, or 64K.
- xv. 4 and 6 only
- xvi. HLEN = 5 (0101 binary), because of minimum 20 bytes header ($5 * 4 \text{ bytes} = 20 \text{ bytes}$)
- xvii. In flooding, the receiving device, forwards the data to all the interfaces, excluding the receiving interface. While in broadcasting, the data is forwarded to all the interfaces including incoming interface.
- xviii. In a very small AS, is it more efficient to use a single OSPF area because the overhead of multiple areas is unnecessary and adds complexity?
- xix. The connection is persistent, but after some time of inactivity, the connection is terminated.
- xx. Prefix: destination being advertised
- xxi. Algorithm complexity is the number of computations/iterations in an algorithm while Message complexity is the number of overhead messages that needed to be transmitted to establish/maintain routes.
- xxii. Application layer.
- xxiii. Due to port # manipulation by network-layer device
- xxiv. No. The packets can be lost in the network. The TCP ensures reliable data transfer by retransmitting the lost packets.

Q2. $6 * 4 = 24$ **(a) 4 marks**

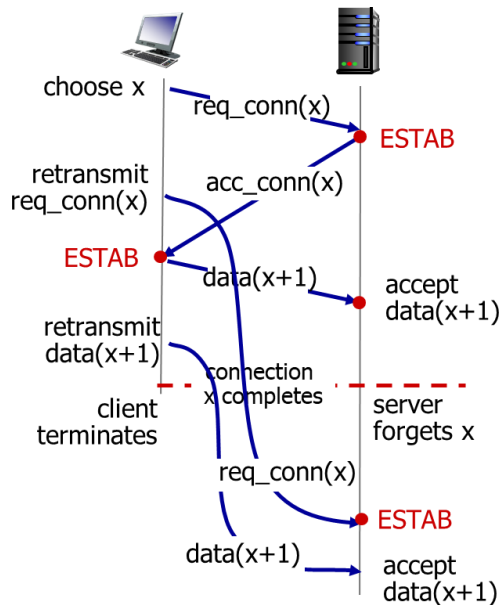
The answer will be like below, but contains some header fields' values, like port#s, IPs etc. instead of just H.



(b) 4 marks

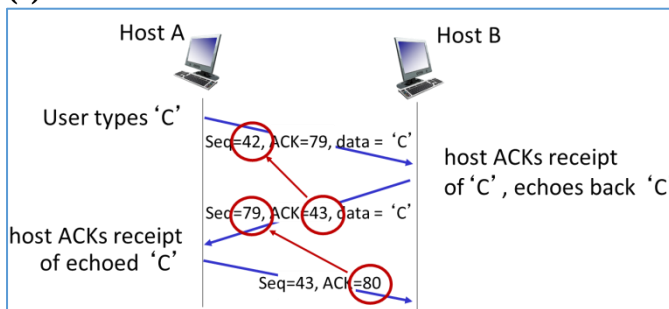


Problem: half open connection! (no client)



Problem: dup data accepted!

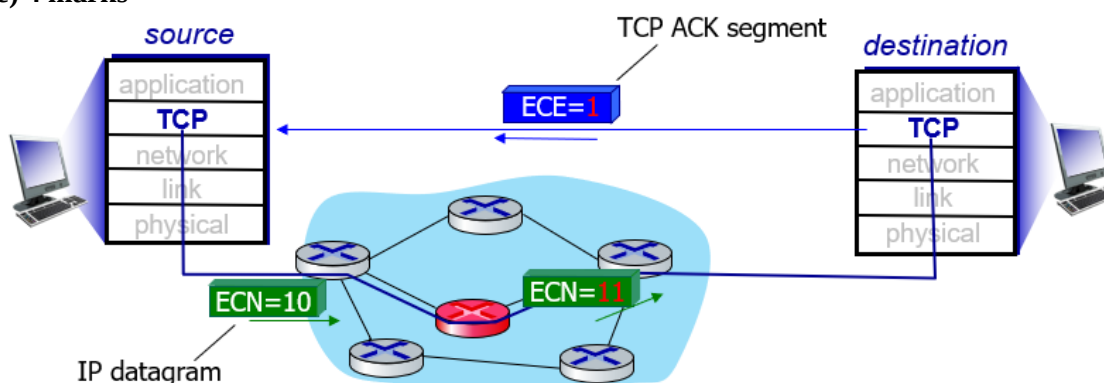
(c) 4 marks



(d) 4 marks

	send window size	receive windows size
Go-Back-N	31	1
Selective-Repeat	16	16

(e) 4 marks



(f) 4 marks

Offset = 100 $\rightarrow$ starting byte = $100 \times 8 = 800$

HLEN = 5 $\rightarrow$ header = 20 bytes

Total length = 100 $\rightarrow$ data = 80 bytes

First byte = 800

Last byte = $800 + 80 - 1 = 879$

Q3. 6 * 4 = 24 marks

(a). 4 marks

- The load balancer provides application-layer routing. It
 - (i) receives external client requests,
 - (ii) directs workload within data center, and distributes the task among multiple servers, while keeping data as close as possible (e.g., in same rack or nearby rack) to minimize tier-2, tier-1 communication.
 - (iii) returns results to external client (hiding data center internals from client)
- It works on the application layer.

(b) 4 marks

The southbound API provides interface for communication between SDN controller and network switches.

The three classes of OpenFlow messages:

- controller-to-switch
- asynchronous (switch to controller)
- symmetric (misc.)

(c) 4 marks

Why divide the Internet into ASes? (Two reasons)

Scalability: Global routing becomes manageable by dividing it into smaller, independently managed networks.

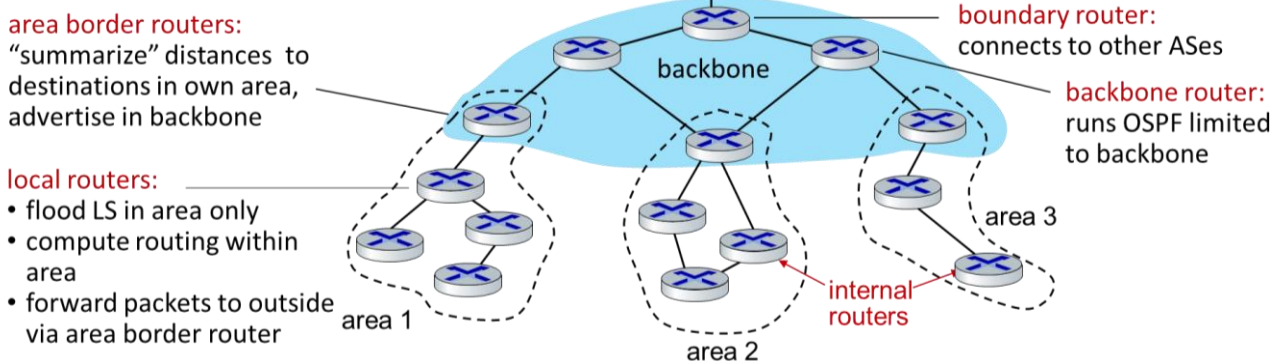
Administrative control: Each AS can set its own routing policies and manage its internal network independently.

Difference in goals:

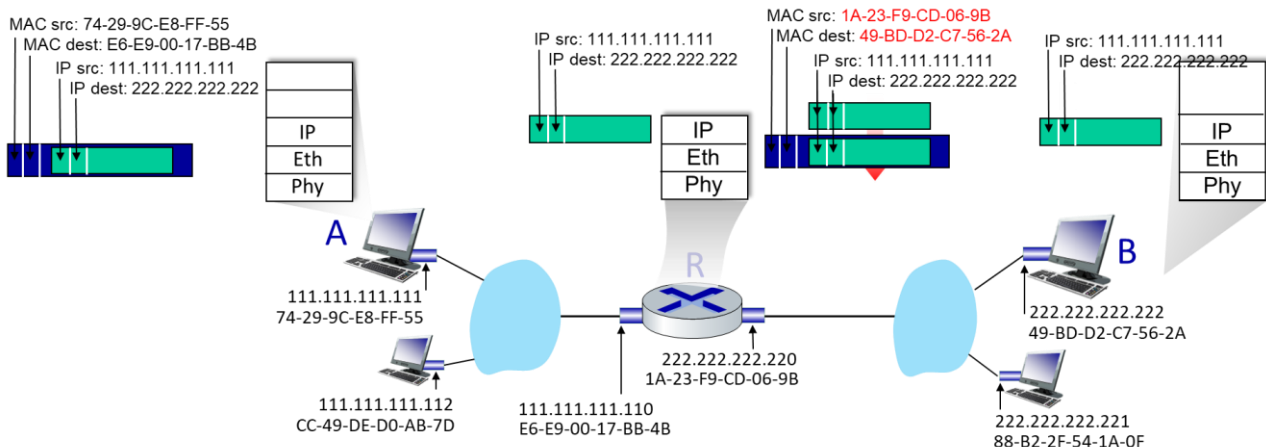
Intra-AS (OSPF): Focus on fast, efficient, and optimal routing within one AS.

Inter-AS (BGP): Focus on policy-based routing and controlling how traffic enters or leaves different ASs.

(d) 4 marks



(e) 4 marks



(f). 4 marks

- The VLAN enables:
 - i. users to physically attach to some other domain's switch, while remains logically attached to their original domain.
 - ii. to divide a subnet into multiple broadcast domains, that helps in scalability, security and efficiency issues.
- Trunk port is the switch port that carries frames between VLANs defined over multiple physical switches.
- 802.1q

Q4. [2 + 3 + 3 + 6 + 6 + 8 = 28 Marks]

(a) 2 marks

-MAC addresses are portable, thus can move interface from one LAN to another.

-While IP address is not portable: it depends on IP subnet to which node is attached

(b) 3 marks

- i. 2
- ii. 7
- iii. 7

(c) 3 marks

Combining several contiguous IP networks into a single larger prefix.

For example: 192.168.10.0/26, 192.168.10.64/26, 192.168.10.128/26, 192.168.10.192/26 can be aggregated into 192.168.10.0/24

(d) 6 marks

Routing Protocol	Routing metric	Update strategy
RIP	Hop count	Sharing periodic updates with neighbor nodes only.
OSPF	Cost (based on link bandwidth)	Event-driven updates; only changes are flooded
BGP	Path attributes (e.g., AS-PATH)	Incremental updates; sent only when routes change, and according to policy

(e) 6 marks

MAC protocol	Advantage	Disadvantage
Slotted ALOHA	single active node can continuously transmit at full rate of channel, simple	collisions, wasting slots, idle slots, clock synchronization
CSMA/CD	efficient at low load, reduces the amount of time wasted in collisions. transmission aborted on collision detection	high load: collision overhead, Exponential backoff selection
Polling	Best channel utilization, No collisions	single point of failure (controller), Polling each node one by one, is inefficient in low load

(c) 8 marks

Device	match	action
Router	longest destination IP prefix	forward out a link
Switch	destination MAC address	forward or flood
Firewall	IP addresses and TCP/UDP port numbers	permit or deny
NAT	IP address and port	rewrite address and port